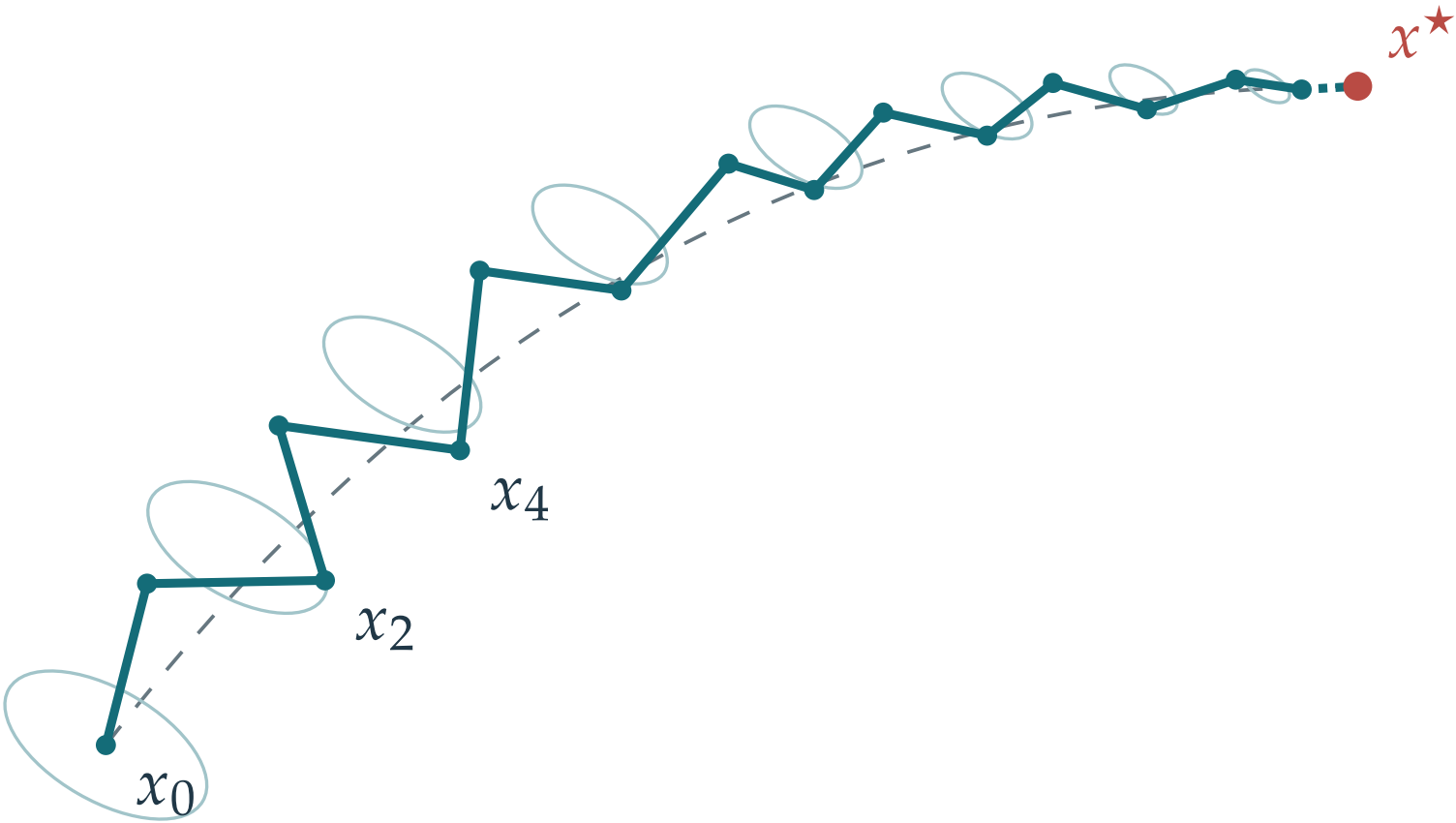


$$x_{k+1} = x_k - \tau_k \nabla f_{i(k)}(x_k)$$

Illustrative trajectory
and shrinking neighborhoods